INbreast Tumor Segmentation with U-Net and Contrastive Learning: Fuzzy Boundaries in Breast Cancer Lesions (Benign vs. Malignant) Using the INbreast Dataset

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1. Modeled deep learning architecture and pipeline

The report outlines a hybrid segmentation pipeline for full-field digital mammograms from INbreast. Its design begins with supervised and self-supervised contrastive learning ideas, then moves toward a Transnet-style architecture that combines a convolutional front end, a transformer encoder, and a U-Net-like decoder. In the correlated github repository, the encoder down samples a 256×256 single-channel mammogram through stacked convolutions into a 32×32 latent map, then applies a transformer encoder to model long-range relations across spatial tokens. The decoder up samples those context-rich features back to pixel space to produce a two-class segmentation map (background versus tumor). This architecture is well matched to mammography because lesion boundaries are often weak, spiculated, or partially occluded by dense tissue; local convolution captures texture and edge fragments, while self-attention helps infer the broader anatomical context around ambiguous borders.

2. Why pixel-level contrastive learning can help

The report's most distinctive idea is dense or pixel-level contrastive learning. Instead of learning one embedding per image, the model learns one embedding per spatial location. Positive pairs are corresponding pixels or boundary pixels drawn from two augmented views of the same case, while negative pairs come from pixels belonging to different tissue types or classes. The intuition is if tumor-boundary pixels cluster together in feature space and background pixels are pushed away, the classifier receives a representation in which class separation is easier. This is especially valuable for fuzzy lesion margins, where standard cross-entropy may learn a blurred decision surface. Contrastive regularization sharpens representation geometry before or during segmentation training, encouraging intra-class compactness and inter-class separation at the level where the clinical task actually operates: the pixel.

3. Bias-variance tradeoff in this method

This pipeline balances bias and variance in several ways. A plain U-Net has stronger inductive bias because convolutions and skip connections strongly assume locality and translation structure; that can improve data efficiency on small medical datasets but may underfit long-range contextual cues. Adding a transformer reduces that architectural bias and increases flexibility, which can lower bias on complex boundary patterns but may raise variance because INbreast is relatively small. Pixel-level contrastive learning partly offsets that variance by giving the encoder a representation-learning objective that uses more supervisory signal per image than a single image label. Data augmentation further reduces variance by exposing the model to multiple plausible boundary-preserving views. Still, variance risk remains high if the model is too large, the train/validation split is small, or lesion-positive cases are rare. The method therefore works best when hybrid structure, moderate capacity, and regularization are used together rather than relying on a large transformer alone.

4. Hyperparameters explicitly present in the notebook

Component	Settings shown in the repository
Contrastive backbone	ResNet-18 skeleton with projector output dimension 128; NT-Xent temperature 0.5.
Dense contrastive block	1×1 convolutional projection head, output dimension 128, L2-normalized features; random pixel subset up to 1024 for memory control.
Segmentation encoder	Input channels 1, hidden dimension 512, transformer layers 4, attention heads 8.
Decoder / output	Transpose-convolution decoder to 256×256; 2 output classes.
Image and view generation	Resize to 256×256; random horizontal flip p=0.5; random vertical flip p=0.5; affine rotation ±15°, translation 10%, scale 0.9–1.1; brightness/contrast jitter 0.3; Gaussian blur kernel 5, sigma 0.1–2.0.
Optimization	Adam optimizer with learning rate 1e-4; batch size 4; fine-tuning loops shown for 3, 5, and 50 epochs, suggesting the longer run is the intended final training stage.

5. Knowledge transfer

To understand and train this model well, we must connect several concepts; Convolution layers act as local linear filters followed by nonlinearities and normalization, so they learn texture detectors and spatial abstractions. The transformer portion applies self-attention, which computes similarity-weighted interactions among spatial tokens; mathematically, this is a learned kernel over patch embeddings. Pixel-level contrastive learning adds another layer of geometry: embeddings are normalized, similarities are usually cosine-like, and the InfoNCE objective increases similarity for matched positives while decreasing it for negatives through a temperature-scaled softmax. For optimization, backpropagation updates encoder, projector, and decoder parameters jointly; good training depends on augmentation design, class imbalance handling, sampling of boundary pixels, and metrics such as Dice and IoU that better reflect segmentation quality than accuracy alone. In this setting, knowledge of morphological boundary extraction, sampling theory, and representation learning is as important as knowing how to code the network itself.

6. Connection to current machine learning and how to improve the pipeline

This approach connects directly to current machine learning in three ways. First, it belongs to the broader shift from purely supervised training toward self-supervised and contrastive pretraining for dense prediction. Second, it aligns with the transformer-to-foundation-model trajectory in medical imaging, where segmentation systems increasingly borrow ideas from promptable and large-scale pretrained vision models. Third, it fits the move toward multimodal medical AI, in which image features are later aligned with reports, pathology, or clinical text. To improve the repository pipeline, I would recommend: replacing placeholder and dummy loops with a reproducible experiment script; adding patient-level rather than image-level data splits; using Dice plus focal or boundary-aware loss together with the contrastive term; sampling hard positives and negatives specifically from lesion borders; calibrating the benign-versus-malignant objective as a joint segmentation-classification task; reporting uncertainty and calibration, not only Dice and IoU; and evaluating whether a modern pretrained medical segmentation foundation model or a fine-tuned SAM/MedSAM-style encoder offers better generalization than a custom encoder trained from scratch. In short, the repository already points in a strong direction: sharper boundary representations, hybrid local-global modeling, and representation learning that uses the limited annotation budget more efficiently.

Selected references

Moreira IC et al., Academic Radiology (2012); Ronneberger O et al., MICCAI (2015); Dosovitskiy A et al., ICLR (2021); Wang X et al., CVPR (2021); Noh S and Lee B-D, Quantitative Imaging in Medicine and Surgery (2025); Ryu JS et al., Biomedical Engineering Letters (2025).